

SPES

Smart Personal Electronic Assistant for the Visually Impaired

A Simple Project Report

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1. What is SPES?

SPES is a small, smart device that helps blind and visually impaired people “read” the world using sound. When the user speaks a command, the device takes a picture of whatever is in front of them — a book, a notice, a signboard or a label — reads the text on it, and speaks the words out loud into the user’s Bluetooth earbuds.

In one line:

SPES turns what you can see into something you can hear.

The user simply says something like “*Jarvis, read this page for me,*” and a few seconds later they hear the text spoken back to them — no screen and no typing needed.

2. Why is it needed?

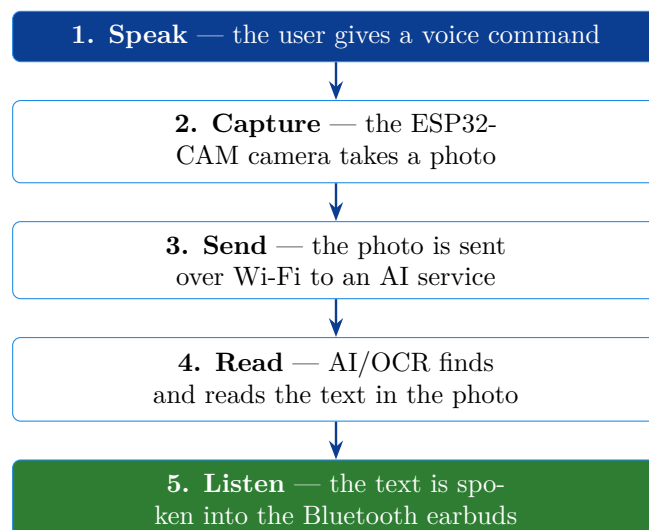
Blind and visually impaired people find many everyday tasks difficult:

- Reading books, letters and documents
- Reading signboards, menus and product labels
- Following printed instructions
- Finding their way in new places

Many existing gadgets that help with this are expensive, complicated, or need a smartphone. SPES aims to be a **simple, low-cost and easy-to-use** helper that works on its own.

3. How does it work?

SPES works in five easy steps:



So the whole idea is very natural: **Speak** → **Capture** → **Process** → **Listen**.

4. What is it made of?

Part	What it does
ESP32-CAM	A tiny camera board that takes the picture
Wi-Fi (built in)	Sends the picture to the internet for processing
Microphone	Listens for the user's voice command
Bluetooth earbuds	Speak the result back to the user
Battery	Makes the device portable
AI / OCR service	Reads the text and turns it into speech

Technologies used: Voice commands, Camera + Computer Vision, OCR (reading text from images), AI, Wi-Fi (IoT), Text-to-Speech, and Bluetooth audio — all working together.

5. Main features

- **Voice activated** — just speak, no buttons to press.
- **Reads text aloud** — books, notices, labels, menus and more.
- **Hands-free** — no need to look at any screen.
- **Portable and wireless** — small enough to carry anywhere.
- **Can guide you** — can later give voice directions to places.

6. How it helps society (SDG)

SPES supports the United Nations Sustainable Development Goals:

- **Main goal — SDG 10 (Reduced Inequalities):** gives blind people equal access to information.
- **SDG 3 (Good Health & Well-Being):** improves independence and quality of life.
- **SDG 4 (Quality Education):** helps blind students read study material.
- **SDG 9 & 11:** uses innovation to make communities more inclusive.

7. Advantages

- Simple and hands-free to use
- Low-cost and portable
- Real-time — gives answers in seconds
- Works without a screen
- Can be improved and extended easily

8. Future improvements

Later, SPES can be made even smarter to also:

- Describe objects nearby (“a chair is in front of you”)
- Warn about obstacles in the path

- Recognise currency notes
- Give full voice-based GPS navigation
- Speak in many languages (English, Tamil, Telugu, Hindi and more)

9. Conclusion

SPES is a simple but powerful idea: use a small camera, the internet and AI to turn text and surroundings into spoken words. This gives visually impaired people more freedom, confidence and independence in their daily life — at a low cost and in an easy, hands-free way.

Prepared by

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